The CASB Oversampling Family

Coverage-Aware Safe-Border SMOTE
Algorithms, evaluation, and a research-paper roadmap

CASB / CASB-relaxed / CASB-edited / CASB-ROS

Document status
Implementation-aligned technical documentation, prepared 20 October 2026. This report describes executed experiments and an experimental method family. It is not a claim that the algorithm is the first of its kind or statistically superior.

Evidence snapshot	Verified scope
Datasets	44 binary KEEL datasets; 42 numeric plus two encoded Abalone datasets
Classifiers	RBF SVM and distance-weighted KNN
Current methods	Nine methods; 7,920 attempted outer evaluations
All-method comparison	40 matched datasets; 26 ADASYN failures across four datasets
Reporting	G-mean, minority F1, precision and recall; evaluated separately
Training selection	Training-only ratio search; no change to models for metric-wise reporting

Prepared from binary_oversampling.py and the executed per-dataset and aggregate notebooks in the project workspace. No experimental result in this report was invented or extrapolated from a paper.

Scope, motivation, and reading map

Binary imbalance occurs when one class has fewer training examples than the other. Ordinary classifiers can favor the majority class; oversampling changes the training distribution by adding minority observations. The central challenge is where to allocate samples and how to generate them without amplifying overlap or isolated noise.

What CASB is intended to do

CASB combines broad minority-source eligibility with extra allocation to locally defined borderline and sparse-region sources. The original method constrains some interpolation steps using distance to the nearest majority example. Later variants relax interpolation, clean majority observations, or replace interpolation with duplication.

These are design motivations, not established guarantees of higher predictive performance. The family can improve some metrics while losing on others, and gains depend on the classifier and dataset.

How the four variants relate

Variant	Source quotas: coverage / border / rare	Generation and cleaning
CASB	50% / 30% / 20%	Conditional step-capped interpolation; no cleaning
CASB-relaxed	70% / 20% / 10%	Unrestricted interpolation; no cleaning
CASB-edited	70% / 20% / 10%	Same relaxed generation, then majority-only ENN
CASB-ROS	50% / 30% / 20%	Duplicate selected sources; no cleaning

Reading map

Sections 2-4 define the common geometry, allocation, and original interpolation. Sections 5-7 explain the variants and selector. Sections 8-12 document evaluation, metrics, results, and interpretation. Sections 13-17 cover reproducibility, a paper outline, references, and the dataset inventory.

Terminology: CASB-SMOTE in the original guide and **CASB** in the benchmark refer to the original interpolation-based variant. **CASB family** refers to the four variants, not to a single classifier or an ensemble of their predictions.

Common geometry and source pools

All distances are computed in the transformed training feature space. Numeric features are median-imputed and standardized. Let M be the minority observations and J the majority observations, with sizes n_min and n_maj. For each minority source x_i, the implementation calculates the following quantities.

Symbol	Exact meaning
N_i	Up to 10 nearest observations in the full training set, excluding the source by index
o_i	Fraction of majority labels in N_i (local overlap)
s_i	Number of minority labels in N_i (local minority support)
d_i	Euclidean distance to the nearest majority observation
K_i	Up to 5 nearest other minority observations
q_i	Distance to the third neighbor in K_i; use the last available neighbor when fewer than three exist

Pool definitions

```
Coverage C = all minority source indices
Border   B = {i: 0.2 < o_i <= 0.7, s_i >= 2, d_i > 0}
Rare     R = {i: q_i >= Q75(q),  s_i >= 2, d_i > 0}
```

Q75(q) is the 75th percentile of the minority-neighbor-distance statistic. The rare pool is a sparse-region heuristic; it is not a fitted cluster detector and does not certify that a source belongs to a meaningful rare subpopulation.

Important implementation details

The pools overlap: every border or rare source is also eligible in the coverage pool, and a source may belong to both border and rare pools. The method samples with replacement, uniformly within each selected pool. Eligibility therefore does not ensure every original source is actually used.

Neighbor exclusion uses observation indices, so duplicate coordinates do not accidentally make an observation its own neighbor. Duplicate minority points can still generate zero-length interpolation steps. Support and majority-distance checks restrict B and R, but not C.

Noise handling is limited: the original method does not delete minority noise. A potentially noisy minority source can still be selected from the coverage pool.

Sample budget, allocation, and fallback

Target ratio

```
rho in {0.50, 0.75, 1.00} in the benchmark
n_new = max(0, ceil(rho * n_maj) - n_min)
```

rho is the desired minority-to-majority ratio before any optional post-generation cleaning. The benchmark returns the original training data when rho is no larger than the current ratio. The CASB implementation uses a ceiling, so its integer target can differ slightly from a library sampler using a different rounding convention.

Integer quotas

For weights w = (w_C, w_B, w_R), first assign floor(w_k * n_new) to each branch. Allocate leftover samples to the largest fractional remainders, with stable coverage-before-border-before-rare tie handling. The branch allocations sum exactly to n_new.

Worked example	Value
Original counts	n_maj = 100; n_min = 20
Selected ratio	rho = 0.75
Required additions	ceil(75) - 20 = 55
CASB / CASB-ROS	Weights (0.5, 0.3, 0.2) give allocations (28, 16, 11)
Relaxed / edited	Weights (0.7, 0.2, 0.1) give allocations (39, 11, 5)
Before cleaning	75 minority observations and 100 majority observations

Empty pools and degenerate cases

If the border or rare pool is empty, its branch draws sources from the coverage pool. The original branch still controls neighbor choice and the step rule: a rare fallback still uses the rare-branch neighbor restriction and margin. This is deliberate reference-implementation behavior, not reassignment of the branch to ordinary coverage.

The sampler requires two classes, finite numeric input and at least two observations per class. Equal class counts return unchanged data. The common benchmark can bypass generation when the selected target ratio is already satisfied.

Optional code path not used in these experiments

CASBSMOTE exposes drop_majority, which can remove a fraction of interior majority observations judged redundant. It is fixed to zero in every reported family experiment. This optional path is different from CASB-edited, which performs ENN after synthesis.

CASB interpolation and its geometric limit

Within a branch, sample source x_s uniformly from its pool and neighbor x_t uniformly from its minority-neighbor list. The rare branch uses only the nearest two minority neighbors when at least two exist. Other branches use up to five.

```
e = x_t - x_s;  L = ||e||
z = x_s + lambda * e
lambda ~ Uniform(0, lambda_max)
```

Condition	lambda_max
Zero-length edge L = 0	0
Coverage branch with o_s <= 0.2	1
Rare branch, including rare fallback	min(1, 0.35 * d_s / L)
Other capped cases	min(1, 0.50 * d_s / L)

Thus, safe-coverage sources can interpolate across the whole selected edge. Other eligible cases are limited relative to their nearest-majority distance. The output keeps the original samples and adds the requested number of minority-labeled points.

Illustrative geometry only; not an observed dataset or empirical result.

A limited deterministic property

For a capped step of length r no larger than c*d_s, with c at most 1/2, every observed majority point b satisfies **||z-b|| >= d_s-r >= r = ||z-x_s||**, by the triangle inequality. The generated point is at least as close to its source as to any observed majority point.

> This elementary bound does not establish the true class of z. It does not apply to unrestricted safe-coverage steps or to the relaxed/edited variants. It is not claimed as a new theorem or a generalization guarantee.

CASB-relaxed and CASB-edited

CASB-relaxed

Relaxed changes two settings simultaneously: source quotas become 70/20/10 and unrestricted=True permits a full interpolation step for every nonzero edge. Pool definitions and rare-branch nearest-two neighbor selection remain unchanged. Original observations are retained.

The motivation is to increase broad coverage and reduce excessively small synthetic steps. The tradeoff is loss of the original step-cap restriction; synthetic samples can extend further into overlap regions. A direct CASB-versus-relaxed comparison cannot isolate whether a change comes from quotas or from disabling caps.

CASB-edited

Edited first generates exactly the same relaxed synthetic set. It then calls EditedNearestNeighbours with **sampling_strategy=[original majority label], n_neighbors=3, kind_sel="mode"**. A targeted majority observation is removed when the three-neighbor majority vote disagrees with its label.

```
Generate using CASB weights (0.7, 0.2, 0.1), unrestricted=True
Apply ENN to the resulting data, targeting ORIGINAL majority only
Retain every original minority observation and every synthetic minority
Raise an error if fewer than two original-majority observations remain
```

The original majority label is retained as the cleaning target even if generation produces equal class counts. ENN is evaluated on the augmented set, so newly synthesized minority observations can influence which majority observations are removed.

Why the final ratio can change

If 75 minority and 100 majority observations exist before ENN, and ENN removes 20 majority observations, the final ratio is 75/80 = 0.9375. The selected ratio 0.75 refers to pre-cleaning counts, not the final dataset. The implementation records removals and final counts in diagnostics.

Removing SMOTEENN from the comparison does not remove ENN from CASB-edited.
CASB-edited still uses its specified cleaner. The standard SMOTEENN baseline has been excluded at the user's request, but a matched-cleaner control remains important for a paper.

CASB-ROS and modular generation

CASB-ROS holds the original CASB geometry, source pools and 50/30/20 quotas fixed, but replaces interpolation with duplication: **z = x_s**. This changes the distribution of repeated source observations compared with ordinary ROS, whose selection is not guided by the three CASB pools.

Property	Ordinary ROS	CASB-ROS
New feature values	None: duplicates existing samples	None: duplicates selected sources
Source preference	Random minority oversampling	Mixture of coverage, border and rare pools
Role of majority geometry	No explicit pool geometry	Used to construct source pools
Synthetic step cap	Not applicable	Not applicable to the duplicated point
Cleaning	None	None

What stays unchanged in code

The implementation still draws a neighbor and interpolation value before setting lambda to zero. These draws have no effect on the duplicated coordinates; they keep the original sampling sequence structure. Audit records therefore include a neighbor and a computed step limit even though the actual point equals its source.

Connection to the uploaded UGO paper [7]

The paper places alternative generators inside an uncertainty-guided framework. Its UGO variants keep their framework and exchange ROS, SMOTE, RWO or MCSMOTE. Our extension borrows this experimental organization: keep CASB source selection and swap generation. We do not implement UGO uncertainty scoring, iterative class-uncertainty testing, adaptive stopping, or noise relocation.

The paper's MCSMOTE denotes a transition-matrix Monte Carlo oversampler, not a multiple-classifier ensemble. MCSMOTE and CASB-MCSMOTE were explicitly skipped. CASB-ROS is also not a 50/50 sequential ROS-plus-SMOTE mixture.

Checks actually performed

A synthetic fixture verified exact source duplication, requested output count, source-quota allocation, preservation of original observations, and identical output for a repeated seed. All 880 CASB-ROS outer evaluations completed across 44 datasets.

Pseudocode and family selection

```
CASB_FAMILY_RESAMPLE(X, y, variant, rho, seed)
  identify minority and original majority labels
  if balanced or target ratio already met: return X, y
  compute overlap, support, minority neighbors, majority distances
  build coverage, border, rare pools
  choose weights: 50/30/20 or 70/20/10
  compute n_new and allocate integer branch quotas
  for each allocated branch draw:
    use coverage sources if that branch pool is empty
    draw source; draw branch-appropriate minority neighbor
    CASB: apply conditional interpolation limit
    relaxed/edited: interpolate on the full edge
    CASB-ROS: duplicate the selected source
    append minority-labeled point and audit record
  if edited: run original-majority-only ENN
  return augmented (and optionally cleaned) training data
```

Training-only selector: a separate experiment

The reusable selector evaluates CASB, CASB-relaxed and CASB-edited at ratios 0.5, 0.75 and 1.0 on the same inner training/validation split. It maximizes the validation mean of G-mean and F1; exact ties prefer a smaller ratio, then CASB, relaxed, edited. The chosen configuration is refitted on the entire outer training fold and assessed on its outer test fold.

This selector does not include CASB-ROS. It was evaluated retrospectively using already-saved inner trials and their corresponding outer refits, then independently reproduced on one numeric and one encoded Abalone setting. Its average selection score was 75.534% across the original 44-dataset study. That historical selection-score result is not the metric-wise family envelope reported later in this document.

Three different meanings of a family result

Comparison	How the variant is chosen	Permissible interpretation
Fixed variant	One predefined variant	Performance of that variant
Training-only selector	Inner validation for each training setting	Performance of a selectable procedure
Retrospective family envelope	Best test average for each metric or dataset	Exploratory potential of the family, not one deployable method

The selector is conventional model selection, not an independently established novel mechanism. Its nine inner trials exceed the three ratio trials used for a fixed baseline.

Evaluation protocol and data handling

Setting	Executed configuration
Outer evaluation	Five supplied KEEL train/test pairs per dataset
Seeds	42 and 137; repeat stochastic resampling and inner splitting on the supplied folds
Inner validation	Stratified 25% holdout taken only from outer training data
Ratio candidates	0.50, 0.75, 1.00; select by validation mean G-mean/F1
Preprocessing	Fit imputation and scaling inside each inner or outer training fit
SVM	RBF kernel, C=1, gamma=scale; no class weighting
KNN	Distance weighting; k=min(7, resampled training count)
Reporting unit	Mean of ten fold/seed scores for each dataset, classifier and method
Aggregation	Equal dataset weight; equal SVM/KNN weight for combined results

Avoiding preprocessing leakage

Each validation trial fits a fresh median imputer and StandardScaler on the inner training subset. The final outer fit repeats preprocessing on all outer training data and transforms the test data without fitting on it. Resampling happens only on the transformed training set.

For Abalone19 and Abalone9-18, Sex is handled by training-fitted most-frequent imputation and dense one-hot encoding with unknown categories ignored. Numeric columns are imputed and standardized separately. Interpolation can create fractional one-hot values; this is an encoded-feature-space experiment, not categorical-value synthesis.

Minority definition and no-op behavior

The minority label is determined from the outer training fold and encoded as 1; metrics target this class. The common sampler wrapper returns unchanged training data when a requested ratio is already met. KNN uses fewer than seven neighbors only when the resampled training set is too small, under the same feasibility rule for all methods.

> The 44 datasets have already informed development or analysis. The 38 datasets outside the first six are a separate reporting cohort, but are not now a pristine, unseen final evaluation set. Supplied folds and related dataset variants also limit independence.

Metric definitions and comparison rules

Metric	Definition for the designated minority class
Recall	TP / (TP + FN): fraction of actual minority examples detected
Precision	TP / (TP + FP): fraction of minority predictions that are correct
F1	2 TP / (2 TP + FP + FN): balances minority precision and recall
Specificity	TN / (TN + FP): majority-class recall
G-mean	sqrt(recall * specificity): balances the two class recalls

Precision and F1 use zero_division=0 in the implementation. The report expresses scores as percentages. Each fold's nonlinear metrics are computed first and then averaged; F1 is not reconstructed from average precision and recall.

The user-requested metric-wise family rule

```
For metric m and each method t:
  A(t,m) = equal-weight dataset/classifier average of metric m
  F(m)   = max A(t,m) over the four CASB variants
  B(m)   = max A(t,m) over SMOTE, BSMOTE, ROS, ADASYN
  family win if F(m) > B(m); ties are reported separately
```

This comparison can select a different fixed family member for each metric. It does not average per-dataset maxima. By contrast, a dataset-wise family count chooses the best variant separately within each dataset and metric; that more flexible result is reported explicitly as retrospective.

Why there are two dataset denominators

There were 7,920 attempted evaluations in the current nine-method set. ADASYN failed in 26 outer settings across four datasets. The all-method comparison excludes those four datasets for every method, yielding 40 matched datasets and 7,200 successful evaluations. The full-coverage comparison without ADASYN retains 44 datasets and 7,040 evaluations.

Excluded from the all-method table: ecoli-0-1-3-7_vs_2-6, iris0, shuttle-c0-vs-c4, shuttle-c2-vs-c4.

> RUS, SMOTEENN and MCSMOTE are not current comparison baselines. No resampling is retained as a reference, but is not included in the four-baseline family-win rule. It can have higher precision than any oversampler.

Verified results: combined and SVM

Equal average of SVM and KNN - 40 matched datasets

Method	G-mean	F1	Precision	Recall
No resampling	62.505	57.504	68.296	52.939
SMOTE	82.160	65.581	61.203	76.777
Borderline-SMOTE	80.777	65.148	59.855	75.708
ROS	82.155	65.674	60.966	76.564
ADASYN	82.349	63.947	57.955	78.199
CASB	81.291	66.316	62.590	75.340
CASB-relaxed	81.887	65.875	61.682	76.121
CASB-edited	83.021	64.755	58.626	79.564
CASB-ROS	81.004	65.940	62.246	74.956

SVM only - 40 matched datasets

Method	G-mean	F1	Precision	Recall
No resampling	59.194	54.653	66.558	49.652
SMOTE	81.619	66.608	63.378	75.738
Borderline-SMOTE	79.669	65.610	61.503	74.052
ROS	82.620	66.951	63.344	76.906
ADASYN	81.499	64.737	59.979	76.397
CASB	80.484	67.276	65.240	74.435
CASB-relaxed	81.376	67.108	64.257	75.203
CASB-edited	82.638	66.102	61.197	78.510
CASB-ROS	80.964	67.444	65.313	75.003

All values are percentages. Each table uses exactly the same 40 datasets across every method. The no-resampling reference is included for context, not as one of the four baselines in the family-win claim.

Verified results: KNN and family envelope

KNN only - 40 matched datasets

Method	G-mean	F1	Precision	Recall
No resampling	65.816	60.354	70.035	56.225
SMOTE	82.702	64.554	59.028	77.816
Borderline-SMOTE	81.884	64.685	58.208	77.364
ROS	81.691	64.396	58.588	76.221
ADASYN	83.200	63.157	55.930	80.001
CASB	82.098	65.355	59.939	76.244
CASB-relaxed	82.399	64.642	59.108	77.038
CASB-edited	83.404	63.408	56.056	80.618
CASB-ROS	81.044	64.436	59.180	74.908

Combined family envelope - strongest average per metric

Metric	Family winner	Score	Best baseline	Score	Gain pp
gmean	CASB-edited	83.021	ADASYN	82.349	0.672
f1	CASB	66.316	ROS	65.674	0.642
precision	CASB	62.590	SMOTE	61.203	1.387
recall	CASB-edited	79.564	ADASYN	78.199	1.365

The family has a higher average on all four requested metrics than each of the four current baselines. This also holds separately for SVM and KNN. For SVM, edited leads the family in G-mean/recall and CASB-ROS leads in F1/precision. For KNN, edited leads in G-mean/recall and CASB leads in F1/precision.

The smallest classifier-specific lead is SVM G-mean: CASB-edited exceeds ROS by approximately 0.018 percentage points. A numerical lead, especially one this small, is not evidence of statistical significance. Different variants supply the four winning averages.

Dataset wins and generator-specific evidence

Family wins on individual datasets - each metric separately

Metric	Wins	Ties strongest	Losses	Datasets
gmean	20	1	19	40
f1	20	1	19	40
precision	24	1	15	40
recall	19	3	18	40

For these counts, any one family member may win a dataset on the specified metric. Different datasets and metrics may select different members. This is distinct from one fixed variant winning or the training-only selector choosing it.

Paired generator comparisons

Comparison	Metric	Mean gain pp	Dataset W / T / L
CASB vs SMOTE	gmean	-0.869	23 / 1 / 16
CASB vs SMOTE	f1	+0.735	24 / 1 / 15
CASB vs SMOTE	precision	+1.387	22 / 2 / 16
CASB vs SMOTE	recall	-1.437	17 / 4 / 19
CASB-ROS vs ROS	gmean	-1.151	16 / 1 / 23
CASB-ROS vs ROS	f1	+0.267	20 / 1 / 19
CASB-ROS vs ROS	precision	+1.280	22 / 1 / 17
CASB-ROS vs ROS	recall	-1.608	11 / 8 / 21

CASB-ROS does not improve every metric over ordinary ROS. Duplication preserves source coordinates but can concentrate training weight on unhelpful sources and cannot fill gaps between observations. The experiments must retain these negative findings.

What changed from earlier counts

Earlier project summaries used a mean G-mean/F1 reporting score and a different baseline list including RUS. Those historical counts should not be mixed with the present ADASYN-inclusive, metric-wise comparisons. This report deliberately declares no overall composite winner.

Reproducibility and implementation use

Repository artifacts

Artifact	Role
binary_oversampling.py	Shared samplers, preprocessing, ratio selection, validation selector, and notebook checkpoint functions
01_... through 44_..._CASB.ipynb	Executed dataset code, per-classifier tables, fold evidence, explicit ADASYN errors
Final_SVM_KNN_Comparison.ipynb	Executed metric-wise matched averages, family comparisons, and full 44-dataset reference
CASB_Selection_and_Novelty_Audit.ipynb	Earlier training-only three-variant selector and preliminary related-work audit

Minimal use on already-preprocessed training arrays

```
from binary_oversampling import CASBSMOTE, CASBEdited

base = CASBSMOTE(ratio=0.75, random_state=42)
relaxed = CASBEdited(sampling_strategy=0.75, clean=False,
                     random_state=42)
edited = CASBEdited(sampling_strategy=0.75, clean=True,
                    random_state=42)
ros = CASBSMOTE(ratio=0.75, generator="ros", random_state=42)

X_resampled, y_resampled = base.fit_resample(X_train, y_train)
```

Fit preprocessing on training data before using these sampler classes. Never resample test data. For the existing benchmark, run all cells in a dataset notebook; current_dataset_records(..., rerun=False) reuses its saved real fits. Set rerun=True to regenerate the current nine methods. This writes checkpoints to the notebook, not progress.md.

Diagnostics and validation

audit_ records source/neighbor indices, interpolation value and limit, overlap, minority support, majority distance, requested branch and actual fallback branch. diagnostics_ records requested/accepted counts, pool sizes, source use and allocations. Edited also exposes removed_indices_ and final class counts. The final notebook verifies record coverage, chosen validation ratios, data hashes, confusion-derived F1/G-mean, and explicit failure handling.

Current report runtime: numpy 2.5.4, scikit-learn 1.9.1, imbalanced-learn 0.14.2.

Implementation SHA-256: d111940b99c8a7b50ecfbf72dfc37f3a2178c31267636ab27ca951a1ad8d3f7d

Writing a paper about the entire family

A coherent paper, rather than four unrelated algorithms

Present a common source-allocation framework and explain how generation constraints and postprocessing produce four variants. A provisional title is **Coverage-Aware Source Allocation for Binary Imbalanced Learning: A Study of Constrained, Relaxed, Edited and Replication-Based Oversampling**. Use this as a working title, not a certified novelty claim.

Paper section	What to include
Introduction	Concrete limitations of uniform interpolation; precise research questions and bounded contributions
Related work	SMOTE, Borderline-SMOTE, ADASYN, safe-level/geometric sampling, local source-type allocation, cleaning, and UGO
Methods	Shared notation, pool definitions, quotas, fallbacks, generation rules, and variant differences
Experimental setup	Data provenance, training-only preprocessing/tuning, matched coverage, budgets, failures, classifiers and metrics
Results	Per-classifier averages, paired generator comparisons, dataset win counts and adverse results
Ablations and sensitivity	Separate quota changes, caps, pool eligibility and cleaner effects
Discussion and conclusion	Limits, family-versus-single-method distinction, and restrained claims

Defensible current result wording

> On 40 datasets with complete evaluations for all methods, at least one CASB variant achieved a higher mean score than each of SMOTE, Borderline-SMOTE, ROS and ADASYN for each of G-mean, minority F1, precision and recall. Different variants attained these maxima. These retrospective family comparisons do not establish that one fixed variant dominates all metrics or that the differences are statistically significant.

Unfinished work required for a stronger contribution

Separate the effect of 70/20/10 quotas from disabling caps; remove border and rare pools one at a time; test a coverage-only version; compare edited CASB with SMOTE followed by the exact same majority-only mode ENN. Add sensitivity to overlap thresholds, neighborhood size and sparse-region cutoff. Freeze choices before a new evaluation on untouched dataset families.

Prior work, limitations, and references

An unpublished method name does not establish mechanism-level novelty. Neighbor-based safety, borderline generation, source-type allocation, geometric synthesis and cleaning have substantial prior art. The exact CASB combination may be a candidate incremental contribution, but its distinctness has not been established by a systematic review.

Potential confounders include related KEEL dataset variants, duplicate feature vectors across supplied folds, reused development datasets, a single inner holdout, only two seeds and two classifiers, unequal search budgets for the selector, and fractional one-hot synthetic inputs. Statistical analysis must not count correlated folds as independent dataset replications. Group-aware duplicate sensitivity and family-aware uncertainty analysis remain to be done.

Selected primary sources and project references

[1] Chawla et al. (2002). SMOTE: Synthetic Minority Over-sampling Technique.
https://doi.org/10.1613/jair.953

[2] Napierala and Stefanowski. Types of minority class examples and their influence on learning classifiers from imbalanced data.
https://doi.org/10.1007/s10844-015-0368-1

[3] Bunkhumpornpat et al. (2009). Safe-Level-SMOTE.
https://doi.org/10.1007/978-3-642-01307-2_43

[4] Douzas and Bacao (2019). Geometric SMOTE.
https://doi.org/10.1016/j.ins.2019.06.007

[5] Koziarski and Wozniak (2024). Local neighborhood encodings for imbalanced data classification.
https://doi.org/10.1007/s10994-024-06563-6

[6] Cawley and Talbot (2010). On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation.
https://www.jmlr.org/papers/v11/cawley10a.html

[7] Saglam and Cengiz (2026). Uncertainty guided oversampling: A classifier-specific approach for effective handling of class imbalance in machine learning. User-supplied PDF; generator framework in Sections 3-4.
https://doi.org/10.1016/j.knosys.2026.115886

[8] imbalanced-learn. Combination of over- and under-sampling.
https://imbalanced-learn.org/stable/combine.html

Project design source: CASB_SMOTE_Detailed_Implementation_Guide.pdf, titled Coverage-Aware Safe-Border SMOTE, prepared 19 October 2026. Earlier Custom_SMOTE_Based_Oversampling_Pipeline.pdf describes a separate predecessor and should not be conflated with the tested CASB family.

Dataset inventory and final evidence boundary

No.	Dataset	40-dataset matched table
1	abalone19	Included
2	abalone9-18	Included
3	ecoli-0-1-3-7_vs_2-6	Excluded: 2 ADASYN failures
4	ecoli-0_vs_1	Included
5	ecoli1	Included
6	ecoli2	Included
7	ecoli3	Included
8	ecoli4	Included
9	glass-0-1-2-3_vs_4-5-6	Included
10	glass-0-1-6_vs_2	Included
11	glass-0-1-6_vs_5	Included
12	glass0	Included
13	glass1	Included
14	glass2	Included
15	glass4	Included
16	glass5	Included
17	glass6	Included
18	haberman	Included
19	iris0	Excluded: 10 ADASYN failures
20	new-thyroid1	Included
21	new-thyroid2	Included
22	page-blocks-1-3_vs_4	Included

The next page completes the inventory. All 44 dataset notebooks have executed current comparison code. Exclusion from matched aggregation does not delete a dataset or its other method results.

Dataset inventory, continued

No.	Dataset	40-dataset matched table
23	page-blocks0	Included
24	pima	Included
25	segment0	Included
26	shuttle-c0-vs-c4	Excluded: 8 ADASYN failures
27	shuttle-c2-vs-c4	Excluded: 6 ADASYN failures
28	vehicle0	Included
29	vehicle1	Included
30	vehicle2	Included
31	vehicle3	Included
32	vowel0	Included
33	wisconsin	Included
34	yeast-0-5-6-7-9_vs_4	Included
35	yeast-1-2-8-9_vs_7	Included
36	yeast-1-4-5-8_vs_7	Included
37	yeast-1_vs_7	Included
38	yeast-2_vs_4	Included
39	yeast-2_vs_8	Included
40	yeast1	Included
41	yeast3	Included
42	yeast4	Included
43	yeast5	Included
44	yeast6	Included

Final reporting boundary: 44 attempted datasets; 40 complete across all nine current methods. Metric-wise family wins are descriptive comparisons of existing test averages. The report does not establish significance, first-in-literature novelty, or universal superiority.